

Project Name: Inventory Management System

1. Project Overview

This project aims to develop an **Inventory Management System (IMS)** that helps businesses track stock levels, orders, sales, and deliveries. The system will automate inventory control and provide real-time reports to minimize stock-related losses.

2. SDLC Phases

Phase 1: Planning

- Identify project objectives and requirements.
- Define key stakeholders (business owners, warehouse managers).
- Estimate budget and project timeline.
- Assess potential risks and mitigation strategies.

Phase 2: System Design

- Design system architecture (front-end: React, back-end: Node.js, database: MySQL).
- Develop ER diagrams and process workflows.
- Plan API endpoints for inventory tracking and user management.

Phase 3: Development

- Implement **user authentication** for secure access.
- Develop inventory **add/update/delete** functionalities.
- Integrate **barcode scanning** for stock updates.
- Build a real-time **inventory dashboard**.

Phase 4: Testing

- Conduct **unit testing** for all modules.
- Perform **integration testing** to validate API interactions.
- Execute **user acceptance testing (UAT)** with warehouse staff.
- Identify and fix performance bottlenecks.

Phase 5: Deployment

- Deploy system to **AWS cloud** with auto-scaling.
- Monitor system logs and performance metrics.

- Provide training materials for end-users.

Phase 6: Maintenance

- Regular security patches and updates.
 - Performance optimization based on user feedback.
 - Feature enhancements for better inventory forecasting.
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3. Business Cost Optimization Insights

- **Planned Budget:** \$40,000
- **Actual Cost:** \$45,000 (Extra expenses on cloud infrastructure).
- **Optimization Suggestions:**
 - Use **serverless computing** to reduce infrastructure costs.
 - Automate routine inventory checks with **AI-based forecasting**.
 - Implement an **API rate limiter** to reduce unnecessary server requests.